

AI Resume-Based Job Matching Agent – Workflow Documentation

Overview

The AI Resume-Based Job Matching Agent is an automation workflow built using **n8n + AI** that analyzes a user's resume, finds relevant job opportunities, and sends curated results via email.

It automates the entire pipeline from:

Resume Input → Skill Extraction → Job Fetching → Matching → AI Explanation → Email Delivery

Objective

To help:

-  Job seekers (especially freshers)
-  Developers building AI tools
-  Professionals exploring automation

Automatically:

- Find relevant job opportunities
 - Match jobs based on skills
 - Generate personalized explanations
 - Deliver results directly via email
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Workflow Architecture

```
Webhook → Set Node → AI Resume Parser → HTTP Request (Jobs API)
→ Job Matching (Code) → AI Explanation → Merge → Email Formatter → Email Node
```

Node-by-Node Breakdown

1 Webhook Node (Input Layer)

Purpose: Receives user input

Input Format:

```
{
  "email": "user@example.com",
  "resumeText": "Your resume content"
}
```

2 Set Node

Purpose:

- Clean and structure incoming data
- Extract required fields

3 AI Resume Parser

Purpose: Extract structured data from resume

Output:

```
{
  "skills": [],
  "roles": [],
  "experience_level": ""
}
```

4 HTTP Request Node (Jobs API)

Service Used: Remotive API

Purpose:

- Fetch job listings based on skills

5 Code Node (Job Matching Logic)

Purpose:

- Compare user skills with job descriptions
- Calculate match score
- Filter top relevant jobs

Logic Includes:

- Skill matching

- Score calculation
 - Sorting + filtering
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6 ✨ AI Agent (Explanation Generator)

Purpose: Generate explanation for why job matches user

Output:

```
{  
  "explanation": "..."  
}
```

7 ✂ Merge Node

Purpose: Combine:

- Job data
- AI explanation
- User email

Mode Used: Append

8 ✎ Code Node (Email Formatter)

Purpose: Format all job matches into a **single structured email**

Example Output:

```
Top Job Matches:  
  
1. Job Title  
Company: XYZ  
Link: ...  
Why: ...  
  
2. Job Title  
...
```

9 ✉ Email Node (Output Layer)

Purpose: Send final results to user

To:

{{json.email}}

Subject:

Your Job Matches 🚀

Input Requirements

Field	Type	Description
email	String	User email address
resumeText	String	Resume content

Output

- Top matched jobs
- Job title, company, link
- AI-generated explanation
- Delivered via email

Key Features

- AI-powered job matching
- Fully automated workflow
- Personalized job recommendations
- Real-time job fetching
- End-to-end automation

Use Cases

- Fresher job search automation
 - Developer job matching tools
 - AI-powered career assistants
 - Automation learning projects
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Limitations

- Depends on job API availability
 - Resume text input only (no file parsing)
 - AI output may vary
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Future Improvements

- Add multiple job APIs
 - Resume file upload support (PDF parsing)
 - Daily job alerts
 - User preference filters
 - Dashboard UI
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Conclusion

This workflow demonstrates how **AI + automation can simplify job searching**.

It serves as a **blueprint for building scalable AI agents using n8n**.

Author

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